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BSIT 2-4
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COMP 009 Object-Oriented Programming

Activity #2

Part I:

1. Create a Java program using bitwise operators that will produce the output: true, 10, false, 15.

Source Code:

```
J New_A2.java 1, U X
School > HW_A2P1 > J New_A2.java > New_A2
1  //package HW_A2P1;
2
3  //Create a Java program using bitwise operators, produces: true, 10, false, 15
4
5  public class New_A2 {
6      public static void main(String[] args) {
7
8          int a = 5;
9          int b = 3;
10
11         boolean result1 = (a & b) == 1;
12         int result2 = a << 1;
13         boolean result3 = (a | b) == 7;
14         int result4 = a ^ 10;
15
16         System.out.println(result1);
17         System.out.println(result2);
18         System.out.println(result3 == false);
19         System.out.println(result4);
20     }
21 }
```

Output:

```
PS C:\Users\Migel\OneDrive\Desktop\Java> cd "c:\Users\Migel\OneDrive\Desktop\Java\School\HW_A2P1\" ; if ($?) { javac New_A2.j
ava } ; if ($?) { java New_A2 }
true
10
false
15
```

2. Create a Java program using the ternary operator that will produce the output:
true, false, 25, 30.

Source Code:

```
J New_A2P1N2.java 1, U X
School > HW_A2P1 > J New_A2P1N2.java > New_A2P1N2
1  //package HW_A2P1;
2
3  //Create a Java program using the ternary operator that will produce
4  //true, false, 25, 30
5
6  public class New_A2P1N2 {
7      Run | Debug
8      public static void main(String[] args) {
9          int x = 10;
10         int y = 20;
11         boolean r1 = (x < y) ? true : false;
12         boolean r2 = (x > y) ? true : false;
13         int r3 = (x < y) ? x + 15 : y;
14         int r4 = (x > y) ? x : y + 10;
15
16         System.out.println(r1);
17         System.out.println(r2);
18         System.out.println(r3);
19         System.out.println(r4);
20     }
```

Output:

```
PS C:\Users\Miguel\OneDrive\Desktop\Java> cd "c:\Users\Miguel\OneDrive\Desktop\Java\School\HW_A2P1\" ; if ($?) { javac New_A2P1N2.java } ; if ($?) { java New_A2P1N2 }
true
false
25
30
```

3. Create a Java program using assignment operators that will produce the output:
Addition: 120, Subtraction: 70, Multiplication: 144, Division: 16.

Source Code:

```
J New_A2P1N3.java 1, U X
School > HW_A2P1 > J New_A2P1N3.java > ...
1  //package HW_A2P1;
2
3  //Create a java program using assignment operators
4  //produces: Addition: 120, Subtraction: 70, Multiplication: 144, Division: 16
5
6  public class New_A2P1N3 {
7      Run | Debug
8      public static void main(String[] args) {
9          int add = 100;
10         add += 20;
11         int sub = 100;
12         sub -= 30;
13         int mul = 12;
14         mul *= 12;
15
16         int div = 64;
17         div /= 4;
18
19         System.out.println("Addition: " + add);
20         System.out.println("Subtraction: " + sub);
21         System.out.println("Multiplication: " + mul);
22         System.out.println("Division: " + div);
23     }
24 }
```

Output:

```
PS C:\Users\Migel\OneDrive\Desktop\Java> cd "c:\Users\Migel\OneDrive\Desktop\Java\School\HW_A2P1\" ; if ($?) { javac New_A2P1N3.java } ; if ($?) { java New_A2P1N3 }
Addition: 120
Subtraction: 70
Multiplication: 144
Division: 16
```

Part II:

Source Code:

```
J New_A2P2N1.java 1, U X
School > HW_A2P2 > J New_A2P2N1.java > New_A2P2N1
1  //Activity #2: Part II
2
3  public class New_A2P2N1 {
4      Run | Debug
5      public static void main(String[] args) {
6          //boolean
7          int m = 7;
8          int n = 4;
9          // 1) Arithmetic + relational
10         boolean output1 = (m * 2 > n + 5);
11         // 1) Logical + Relational
12         boolean output2 = !(m == n) && (m > n);
13
14         //numeric
15         // 1) Bitwise operators
16         int output3 = m << 2;
17         // 1) Assignment operator
18         int p = 30;
19         p += 15;
20         int output4 = p;
21
22         //numeric
23         // 3) tricky increment case
24         int q = 5;
25         int output5 = q++ + ++q;
26         // 6. Arithmetic computation
27         int output6 = (20 - 3) * 2;
28
29         //2) six outputs
30         System.out.println(output1);
31         System.out.println(output2);
32         System.out.println(output3);
33         System.out.println(output4);
34         System.out.println(output5);
35         System.out.println(output6);
36     }
37 }
```

Output:

```
PS C:\Users\Migel\OneDrive\Desktop\Java> cd "
N1.java } ; if ($?) { java New_A2P2N1 }
true
true
28
45
12
34
```